

A Note on Standard Errors in the Hazard-Matching GMM Estimator

This note explains how to obtain asymptotic standard errors for the structural estimates without using the bootstrap. The key point is that the estimator is, in essence, a minimum-distance or GMM estimator, where model-implied moments are matched to empirical moments. The only additional complication is that the empirical moments are themselves estimated hazards rather than raw sample averages. That complication changes the covariance matrix of the moments, but it does not change the general GMM logic.

Let $\hat{m}$ denote the vector of empirical moments and let $m(\theta)$ denote the corresponding vector of model-implied moments for parameter vector θ . The estimator solves

$$\hat{\theta} = \arg \min_{\theta} (m(\theta) - \hat{m})' W (m(\theta) - \hat{m}), \quad (1)$$

where W is a positive definite weighting matrix. This is exactly a minimum-distance estimator and may be viewed as a GMM estimator with moment condition $g(\theta) = m(\theta) - \hat{m}$.

Under standard regularity conditions, the asymptotic covariance matrix of $\hat{\theta}$ is given by the usual sandwich formula

$$\widehat{\text{Var}}(\hat{\theta}) = (G'WG)^{-1}G'W\hat{\Omega}WG(G'WG)^{-1}, \quad (2)$$

where

$$G = \left. \frac{\partial m(\theta)}{\partial \theta'} \right|_{\theta=\hat{\theta}} \quad (3)$$

is the Jacobian of the model-implied moments with respect to the structural parameters, and $\hat{\Omega}$ is the covariance matrix of the empirical moment vector $\hat{m}$.

If the weighting matrix is chosen efficiently as $W = \hat{\Omega}^{-1}$, the expression simplifies to

$$\widehat{\text{Var}}(\hat{\theta}) = (G'\hat{\Omega}^{-1}G)^{-1}. \quad (4)$$

In practice, however, it is often convenient to use a diagonal weighting matrix based on inverse moment variances. In that case, the full sandwich formula should still be used for standard errors. One should not use the efficient-GMM simplification unless the full covariance matrix of the moments is in fact used in the criterion.

The main practical issue is therefore not the GMM formula itself, but how to obtain $\widehat{\Omega}$. There are three natural cases.

The simplest case is when the empirical moments are raw weekly hazards. If at duration t there are n_t individuals at risk and d_t exits, then the empirical hazard is

$$\hat{h}_t = \frac{d_t}{n_t}. \quad (5)$$

A simple approximation treats this as a binomial proportion and uses

$$\text{Var}(\hat{h}_t) \approx \frac{\hat{h}_t(1 - \hat{h}_t)}{n_t}. \quad (6)$$

If one is willing to ignore covariance across durations, one can then let $\widehat{\Omega}$ be diagonal with these entries. This is easy to implement, though clearly somewhat crude because hazards at different durations are mechanically related through the survivor process.

A better case, and probably the one most relevant here, is when the empirical hazards are generated in a first step from a parametric or semiparametric model, for example a logit or cloglog specification with b-splines. In that setting the empirical hazard vector can be written as

$$\hat{m} = g(\hat{\beta}), \quad (7)$$

where $\hat{\beta}$ is the vector of first-step parameters and $g(\cdot)$ maps those first-step parameters into the fitted weekly hazard moments used in the second-step structural estimation. Then the delta method gives

$$\widehat{\Omega} = D \widehat{\text{Var}}(\hat{\beta}) D', \quad D = \left. \frac{\partial g(\beta)}{\partial \beta'} \right|_{\beta=\hat{\beta}}. \quad (8)$$

This is the cleanest analytical route. It says that one should first obtain the covariance matrix of the spline or logit coefficients from the first-step hazard estimation, and then propagate that uncertainty into the weekly fitted hazards through the Jacobian D .

Once $\widehat{\Omega}$ is available, the remaining ingredient is the Jacobian G of model moments with respect to the structural parameters. In a structural model of this kind, G will often be easiest to compute

numerically using finite differences. For the j th parameter,

$$G_{.j} \approx \frac{m(\hat{\theta} + \varepsilon e_j) - m(\hat{\theta} - \varepsilon e_j)}{2\varepsilon}, \quad (9)$$

where e_j is the unit vector in the j th coordinate and ε is a small step size. This is usually straightforward because the estimator already requires repeated solution and simulation of the model.

Putting the pieces together, the non-bootstrap route is therefore as follows. First, estimate the empirical hazard in the first step and obtain the covariance matrix of the estimated hazard moments. Second, compute the numerical Jacobian of the model-implied hazard vector with respect to the structural parameters at the estimated parameter vector. Third, plug both objects into the sandwich formula. The resulting square roots of the diagonal elements give asymptotic standard errors for the structural parameters.

It is useful to emphasize what this approach does and does not require. It does not require resampling or repeated full estimation on bootstrap samples. It does require that the first-step uncertainty in the empirical hazards be accounted for explicitly. If one ignores that first-step uncertainty and treats the empirical hazards as fixed, the resulting standard errors will generally be too small.

In the present application, this means that a valid analytical standard-error calculation is perfectly possible. The estimator is not fundamentally different from any other two-step minimum-distance estimator. The only object that requires some care is the covariance matrix of the estimated hazard profile. If those hazards are based directly on raw exit rates, a binomial approximation may be sufficient as a first pass. If they are based on a spline-smoothed first step, the more principled route is to use the delta method starting from the covariance matrix of the first-step spline estimates.

Finally, there is a practical caveat. Even though the asymptotic formula is standard, it may behave poorly when some structural parameters are weakly identified or sit near the boundary of the parameter space. In such cases the numerical Jacobian can be unstable, and the resulting standard errors may be noisy or misleading. This does not invalidate the formula, but it does mean that the results should be interpreted together with other diagnostics, such as sensitivity to starting values, profile plots of the objective function, and the economic plausibility of the parameter estimates.